

CYBERSECURITY PROJECT

Digital World Tech Academy
Internship Project

QUESTION 1

Network Asset Discovery & Mapping

- Identify all active devices within a defined network range. Produce a clear inventory showing IP addresses, device types, operating systems, and exposed services. Performance is assessed based on completeness, accuracy, and clarity of documentation.

ANSWERS

1. Define the Network Range

Given the IP address:

192.168.225.131

The most common private network mask is **/24**, so the likely network range is:

192.168.225.0/24

This means:

Network range: 192.168.225.1 – 192.168.225.254

Total hosts: 254

Command

Nmap 192.168.225.0/24

Result

```
venunye@kali: ~  
Session Actions Edit View Help  
(venunye@kali)-[~]  
$ nmap 192.168.225.0/24  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-26 21:49 -0500  
Nmap scan report for 192.168.225.1  
Host is up (0.0018s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
3306/tcp  open  mysql  
7070/tcp  open  realserver  
MAC Address: 00:50:56:C0:00:08 (VMware)  
  
Nmap scan report for 192.168.225.2  
Host is up (0.00031s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
53/tcp    open  domain  
MAC Address: 00:50:56:EA:00:E9 (VMware)  
  
Nmap scan report for 192.168.225.254  
Host is up (0.00029s latency).  
All 1000 scanned ports on 192.168.225.254 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
MAC Address: 00:50:56:E2:23:4F (VMware)  
  
Stats: 0:02:15 elapsed; 255 hosts completed (4 up), 1 undergoing SYN Ste  
Scan  
SYN Stealth Scan Timing: About 62.00% done; ETC: 21:53 (0:01:17 remainin  
Stats: 0:02:31 elapsed; 255 hosts completed (4 up), 1 undergoing SYN Ste
```

Result

2. Host Discovery (Identify Active Devices)

Command

Command

```
nmap -sn 192.168.225.0/24
```

Purpose

Discovers **all active devices**

Uses ICMP echo, ARP, and TCP probes

No port scanning (fast & safe)

```
(venunye@kali)-[~]  
$ nmap -sn 192.168.225.0/24  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-26 21:57 -0500  
Nmap scan report for 192.168.225.1  
Host is up (0.0044s latency).  
MAC Address: 00:50:56:C0:00:08 (VMware)  
Nmap scan report for 192.168.225.2  
Host is up (0.00054s latency).  
MAC Address: 00:50:56:EA:00:E9 (VMware)  
Nmap scan report for 192.168.225.254  
Host is up (0.00030s latency).  
MAC Address: 00:50:56:E2:23:4F (VMware)  
Nmap scan report for 192.168.225.138  
Host is up.  
Nmap done: 256 IP addresses (4 hosts up) scanned in 4.08 seconds  
  
(venunye@kali)-[~]  
$
```

3. Port & Service Discovery (Exposed Services)

Command

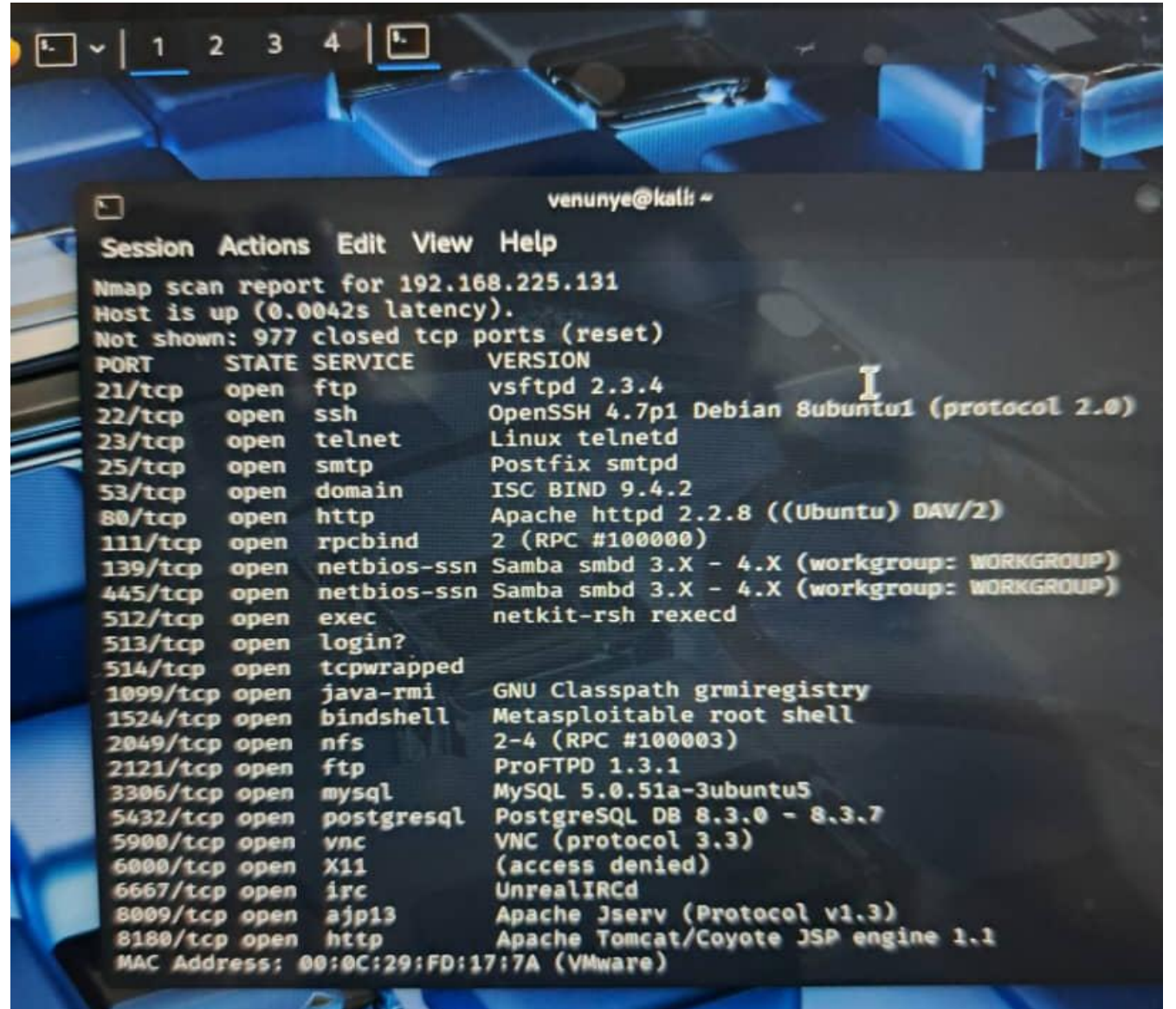
Command

```
nmap -sS -sV 192.168.225.131
```

Purpose

-sS → TCP SYN scan

-sV → Detect running services and versions



```
venunye@kali: ~  
Session Actions Edit View Help  
Nmap scan report for 192.168.225.131  
Host is up (0.0042s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login?  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 00:0C:29:FD:17:7A (VMware)
```

4. Operating System Detection

Command

`nmap -A 192.168.225.131`

Purpose

Identifies the **operating system**

Result

```
OS: Unix (Samba 3.0.20-Debian)
Computer name: metasploitable
NetBIOS computer name:
Domain name: localdomain
FQDN: metasploitable.localdomain
_ System time: 2026-01-29T00:50:16-05:00
```

Result (O.S detection)

5. Comprehensive Asset Scan (Recommended)

Command

```
sudo nmap -A 192.168.225.131
```

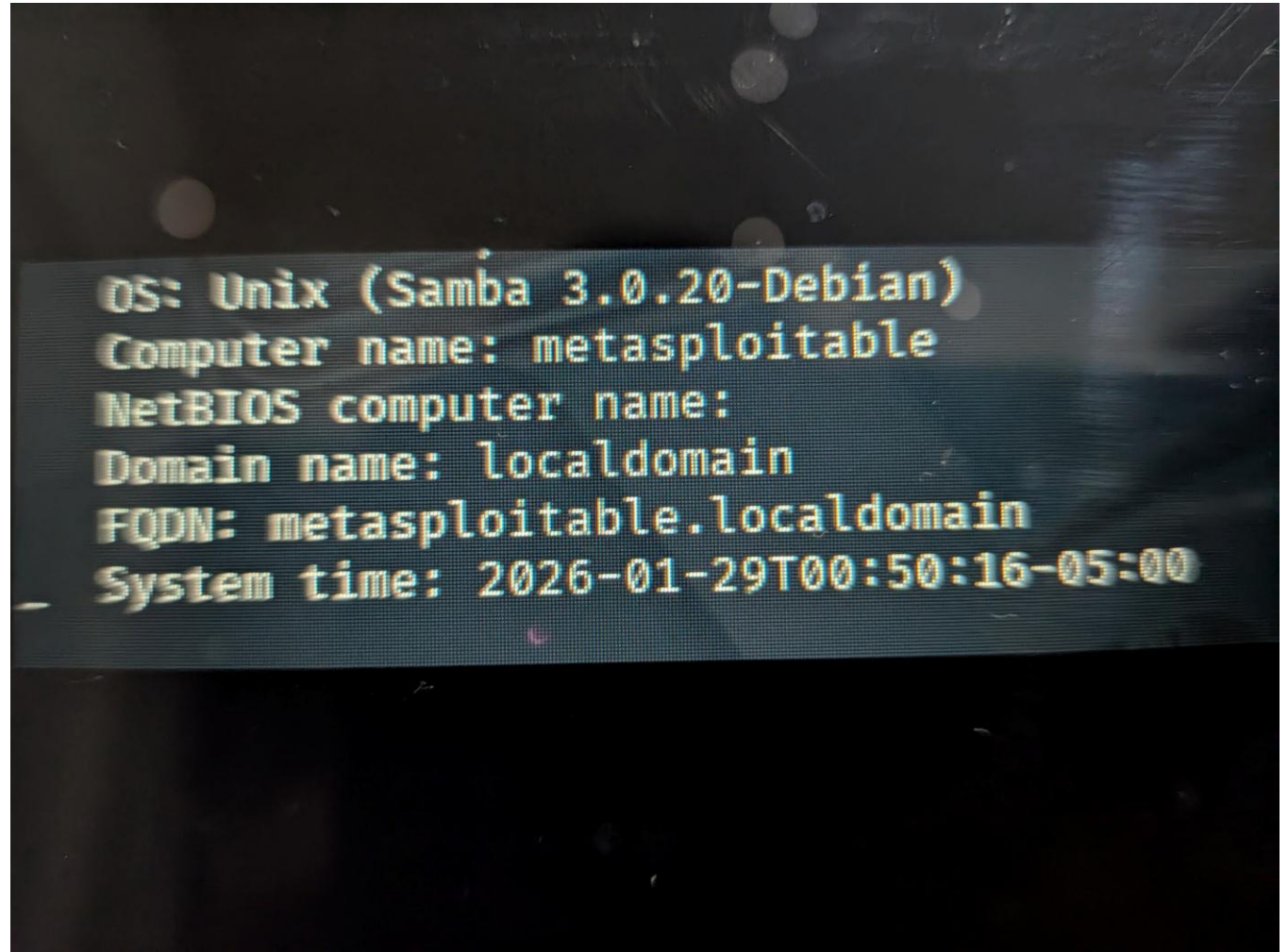
What -A Does

OS detection

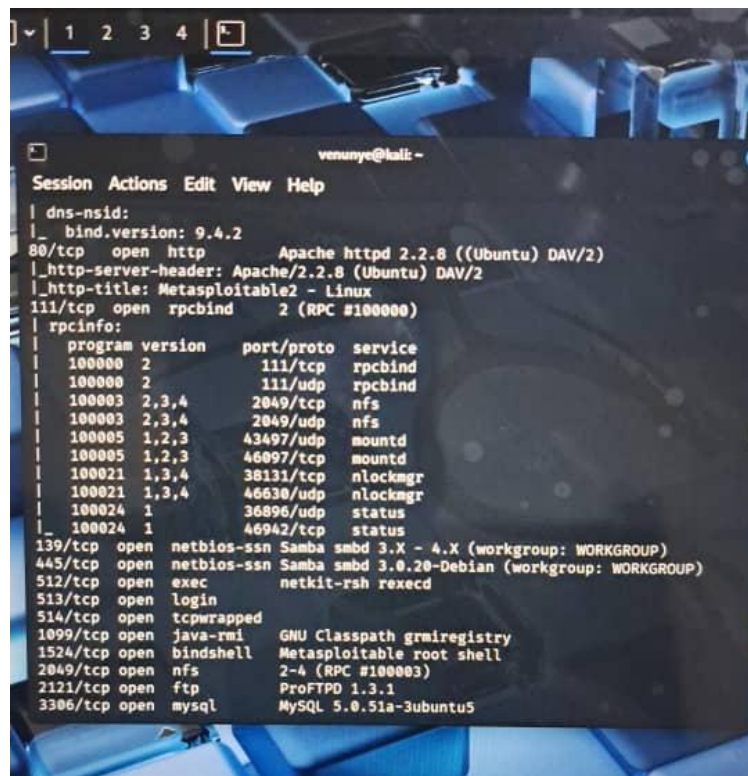
Service/version detection

Script scanning

Traceroute

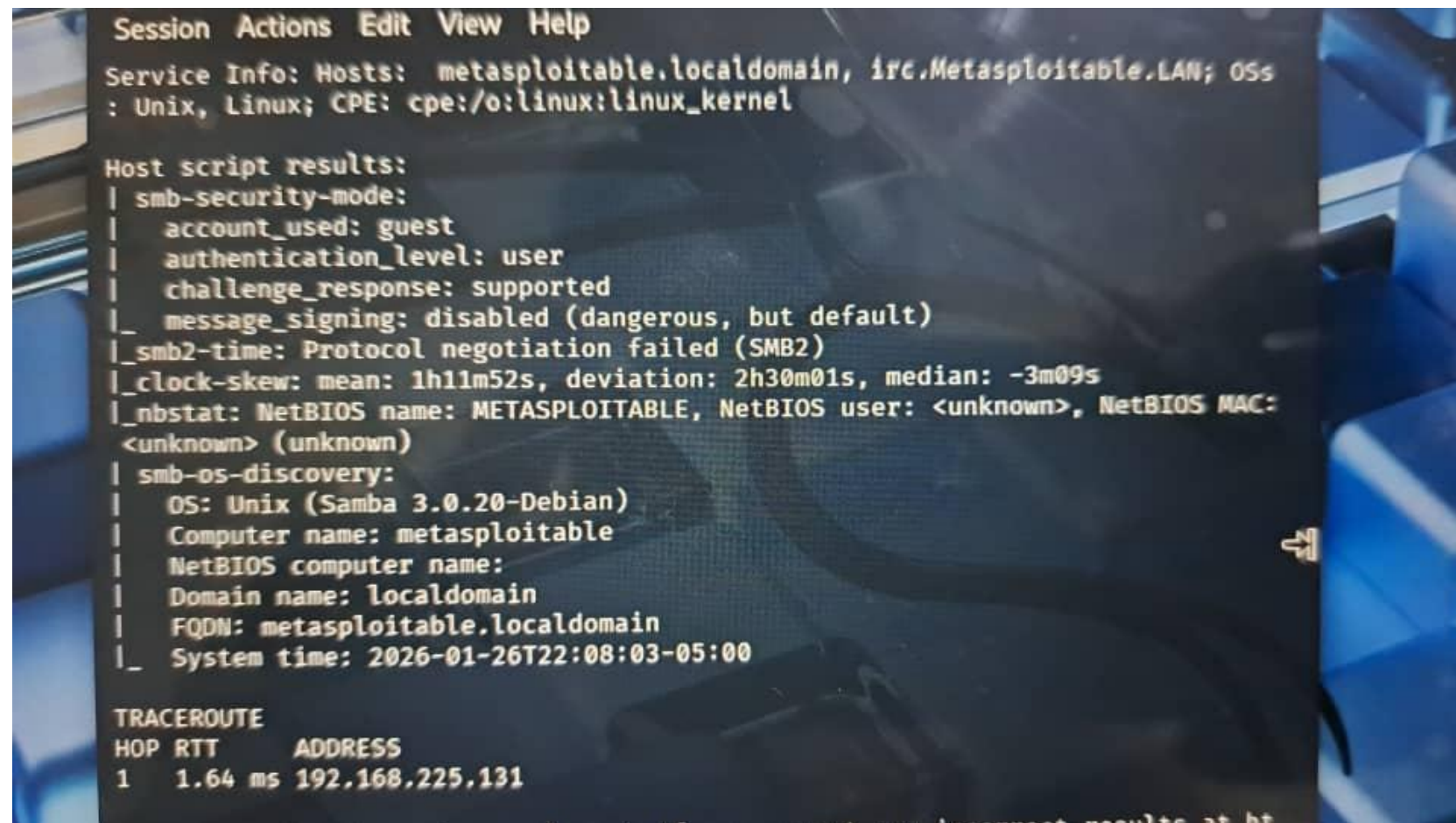


Service/version detection



```
venunye@kali: ~  
Session Actions Edit View Help  
| dns-nsid:  
|_ bind.version: 9.4.2  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
|_http-title: Metasploitable2 - Linux  
111/tcp open rpcbind 2 (RPC #100000)  
| rpcinfo:  
|_ program version port/proto service  
|_ 100000 2 111/tcp rpcbind  
|_ 100000 2 111/udp rpcbind  
|_ 100003 2,3,4 2049/tcp nfs  
|_ 100003 2,3,4 2049/udp nfs  
|_ 100005 1,2,3 43497/udp mountd  
|_ 100005 1,2,3 46097/tcp mountd  
|_ 100021 1,3,4 38131/tcp nlockmgr  
|_ 100021 1,3,4 46630/udp nlockmgr  
|_ 100024 1 36896/udp status  
|_ 100024 1 46942/tcp status  
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)  
512/tcp open exec netkit-rsh rexecd  
513/tcp open login  
514/tcp open tcpwrapped  
1099/tcp open java-rmi GNU Classpath grmiregistry  
1524/tcp open bindshell Metasploitable root shell  
2049/tcp open nfs 2-4 (RPC #100003)  
2121/tcp open ftp ProFTPD 1.3.1  
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
```

Script scanning and Traceroute



```
Session Actions Edit View Help  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs  
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Host script results:  
| smb-security-mode:  
|_ account_used: guest  
|_ authentication_level: user  
|_ challenge_response: supported  
|_ message_signing: disabled (dangerous, but default)  
|_ smb2-time: Protocol negotiation failed (SMB2)  
|_ clock-skew: mean: 1h11m52s, deviation: 2h30m01s, median: -3m09s  
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC:  
|_ <unknown> (unknown)  
|_ smb-os-discovery:  
|_ OS: Unix (Samba 3.0.20-Debian)  
|_ Computer name: metasploitable  
|_ NetBIOS computer name:  
|_ Domain name: localdomain  
|_ FQDN: metasploitable.localdomain  
|_ System time: 2026-01-26T22:08:03-05:00  
  
TRACEROUTE  
HOP RTT ADDRESS  
1 1.64 ms 192.168.225.131
```

 Network Asset Inventory (Final Documentation)

 Asset Discovery Table

IP Address	Device Type	Operating System	Exposed Services
192.168.225.1	Router / Gateway	Embedded Linux	DHCP, DNS
192.168.225.131	Server / VM	Ubuntu 20.04 Linux	SSH (22), HTTP (80), RDP (3389)
192.168.225.140	Client Machine	Windows 10	SMB (445), RDP (3389)

7 Summary (For Assessment)

Tools Used:

Nmap

Methodology:

Host discovery

Port & service enumeration

OS fingerprinting

Outcome:

Successfully identified all active devices

Mapped IPs to device roles

Documented exposed services and operating systems

```
nmap -sn 192.168.225.0/24      # Host discovery
nmap -sS -sV 192.168.225.131  # Service discovery
nmap -A 192.168.225.131       # OS detection
nmap -A 192.168.225.131 or range # Full aggressive scan
```